gendertagger: A Multi-Method Workflow for Gender Annotation in Digital Scholarly Editions

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Abstract: Digital scholarly editions frequently rely on person registers to reconstruct historical networks and contextualize sources. However, gender information in such registers is often incomplete, making structural imbalances within historical datasets difficult to analyze. This paper presents *gendertagger*, a multi-method workflow for semi-automated gender annotation of TEI/XML person registers. The workflow combines authority-data lookup, heuristic rules, dictionary-based methods, probabilistic name classifiers, transformer-based classifiers, and a locally hosted large language model. Rather than treating classification as a fully automated task, the system records provenance, confidence, and inter-method agreement in order to support human editorial review. Using the *schleiermacher digital* person register, comprising 8,277 entries, as a case study, the paper contributes a TEI-compatible modeling strategy, an uncertainty-aware aggregation workflow, and a critical reflection on the methodological limits of computational gender classification in historical corpora.

Keywords: gender data gap, digital humanities, digital scholarly editions

1 Introduction

As critical representations of historical sources [Sah13], digital scholarly editions rely heavily on structured metadata and person registers. Such lists of persons serve to gather information about writers of the source itself, but also about people who are mentioned within the edited documents or who stand in some other relation to them. However, these registers are particularly affected by the gender data gap [CP20], which concerns not only areas such as medicine or the work environment, but also humanities research. Person registers of digital editions frequently reflect a pronounced imbalance between male and female individuals. This imbalance is rooted partly in the structural asymmetry of historical sources and traditions of canon formation, and partly in the limitations of contemporary metadata infrastructures. Authority files provide identifiers and descriptive records for persons and other entities. This study uses the German Integrated Authority File (*Gemeinsame Normdatei*, GND), where gender information is optional in person records [Sta26]. Consequently, identifying a person through GND does not necessarily provide the gender information required for annotation. GND subject terminology can also reflect male-centered and binary assumptions [Ale25].

In most cases, the gender imbalance of digital edition corpora can neither be precisely quantified nor made visible, and editions rarely offer the means to filter or search for persons of a specific gender.[1] One practical reason why gender metadata is so often absent is scale: the increasing size of digital editions makes manual enrichment of person registers a time-consuming and resource-intensive task. Automated approaches to gender classification based on personal names therefore offer promising possibilities not only for supporting editorial workflows, but also for making structural asymmetries within historical corpora computationally analyzable. In the context of digital scholarly editions, moreover, the absence of gender metadata may itself contribute to the invisibility of marginalized historical actors and limit the critical analysis of canon formation and representation within edited corpora.

Automated name-based gender classification has been widely applied across fields such as bibliometrics, computational social science, and digital humanities. Existing approaches range from dictionary- and authority-based systems to statistical and machine-learning methods trained on large name datasets [KWL+16, BCL23]. Comparative evaluations have shown that the performance of such methods varies considerably depending on linguistic and cultural context, the composition of the underlying training data, and the handling of ambiguous names [KWL+16, SM18]. Within the digital humanities, name-based gender classification has been used, for example, to investigate gender distributions in historical corpora and authorship datasets [BM15]. At the same time, research has emphasized that these approaches rely on probabilistic and culturally contingent associations between names and gender categories, raising methodological questions about transparency, uncertainty, and bias [BCL23].

These challenges become especially pronounced in historical corpora. Blevins and Mullen demonstrate that the gender association of names may shift over time and across cultural contexts, making contemporary gender datasets unreliable for many historical sources [BM15]. Automated approaches in historical contexts therefore remain methodologically difficult due to ambiguous names, inconsistent spelling variants, incomplete authority data, and the risk of reproducing historical and algorithmic biases. Beyond these technical limitations, the modeling of gender information is itself a subject of ongoing ethical debate. Scholars have questioned whether inferring gender from names risks reinforcing binary gender assumptions, reproducing cultural biases, or obscuring the distinction between socially assigned and self-identified gender categories [BCL23]. As Van Buskirk et al. note, the literature has not reached a consensus on whether such methods should be used at all, although compelling arguments have been made both in favor of and against their use [BCL23, p. 867]. In the context of current societal debates surrounding gender, computational approaches must therefore navigate carefully between sensitivity to present-day understandings and the pragmatic realities of categorizing historical persons. Recent discussions in digital humanities have emphasized the importance of "ethical data work," which combines critical reflection on data categories and infrastructures with transparent and accountable computational practices [LS26].

Despite the growing body of research on name-based gender classification, comparatively little

[1] One example is the set of biographies in the digital edition *Praktiken der Monarchie*, which allows users to filter entries by gender (female, male, unknown), see Akademienvorhaben Anpassungsstrategien der späten mitteleuropäischen Monarchie am preußischen Beispiel (1786–1918), ed. Berlin-Brandenburgische Akademie der Wissenschaften. Praktiken der Monarchie: Biogramme. Version 9, August 20, 2025. https://actaborussica.bbaw.de/P0006319.

work has addressed how such approaches can be integrated into the editorial workflows, metadata standards, and epistemological requirements of digital scholarly editions. To address these challenges, we developed gendertagger, a multi-method workflow for semi-automated gender annotation of person registers in digital scholarly editions. Such registers are typically encoded in TEI/XML, where TEI refers to the Text Encoding Initiative, a community standard and vocabulary for the description of historical texts.[2] To assign a gender to a person entry, *gendertagger* combines multiple classification approaches: retrieval of recorded gender information from a local GND snapshot using the register's GND identifiers, heuristic methods, statistical models, transformer-based models, and local large language models. These methods are embedded within a human-in-the-loop workflow that preserves editorial oversight and supports uncertainty-aware review processes.

We selected *schleiermacher digital* [SMD12] because it combines a manageable corpus with enrichment challenges relevant to our planned work on other editions at the Academy. Its temporal and regional focus is comparable to that of these editions, while incomplete names, descriptions based on family relationships, and limited authority coverage provide concrete tests of complementary annotation methods. Our collaboration with its editorial team also enabled us to address a practical need: enriching the person register as the project approaches the end of its term. The case study therefore provides an initial setting for developing the workflow and identifying requirements for its subsequent adaptation to other editions.

This paper makes the following contributions: it presents a TEI/XML modeling approach for gender annotations that encodes provenance, confidence, and editorial responsibility; it describes and evaluates a multi-method classification pipeline and the inter-method agreement measures used to triage editorial attention; it quantifies the gender distribution of a digital edition's register to describe its gender data gap; and it reflects critically on the methodological and ethical challenges of computational gender classification in historical corpora.

2 Study Design

The initial aim of our study was to quantify the gender distribution of the *schleiermacher digital* person register.[3] This task required several methodologically and ethically sensitive decisions that shaped the design of our study and are discussed in the following section. The study is implemented as a Python pipeline, together with a small interactive review prototype used to validate selected entries of the generated output. All components run locally, including the large language model, so that no register data leaves the editorial environment, a constraint that projects with restrictive licensing or archival policies often face.

2.1 Register Data and Design Requirements

Person registers of digital scholarly editions can impose a variety of constraints on automated enrichment processes. In our study, five key constraints shaped the design of our approach.

[2] https://tei-c.org/. Accessed 23 September 2026.
[3] https://schleiermacher-digital.de/register/personen/index.xql. Accessed 23 September 2026.

Heterogeneous Register Entries Register entries are heterogeneous in what amount of information they offer: a minority carries identifiers that resolve to structured authority records, while the majority provide only a name string and ideally vital dates or a short biographical note; moreover, some entries refer not to individuals at all but to families, corporate bodies, or married couples. Some identify entities only by initials.

Classifier Coverage and Output Normalization While researching a suitable name-based approach we found that no single classifier covers this spectrum: authority lookup, heuristic name dictionaries, statistical name models, transformer-based classifiers, and locally-hosted large language models each use different data points in their analysis and can differ on principled grounds, e. g. different training corpora or different operationalization of gender. The methods also differ in their raw outputs and can include binary male/female, `mostly_male`, `andy`, `conflict`, `uneindeutig` [ambiguous], `organisation` or a silent answer by an authority record. For a meaningful comparison an explicit normalization onto a shared label set was performed. This normalization is also a methodological limitation, as collapsing method-specific outputs is a reduction and discarding information that the original classifiers expressed.

Limits of Manual Review A manual review of all automatically generated classifications was not possible due to time constraints. Therefore the system has to include a method to track and flag contested records for human quality assurance. In this process methods are checked against one another through inter-method agreement, where the disagreement becomes a primary signal that is used to spot questionable results.

Provenance and Interpretive Responsibility Because gender attribution by an annotator or algorithm who is neither the person classified nor their contemporary is an interpretive act rather than a clear observation, every machine-assigned label must carry explicit provenance and who or what assigned it and with what confidence.

Organizations in the Person Register A peculiarity of the *schleiermacher digital* data was that, due to previous editorial decisions, several organizations had been included in the person register. Our approach had to account for these during the enrichment process and helped to solve this issue in a separate processing step. Today the digital edition includes a separate register for organizations.[4]

2.2 Encoding Gender in TEI/XML

By *gender* we refer to gender categories historically or culturally associated with personal names within available metadata sources, rather than the self-identified gender of an individual. Within many historical datasets, name-based classification remains one of the few scalable approaches currently available for making structural gender imbalances and the gender data gap visible. The classifications generated by *gendertagger* are derived from socially and historically informed name-gender associations [BCL23] found in metadata, authority records, and computational

[4] https://schleiermacher-digital.de/register/institutionen/index.xql

models — more precisely, the system infers the gendered associations that historical sources attach to personal names. We therefore treat the resulting annotations as inferred metadata categories for the analysis of structural representation within historical corpora, not as authoritative claims about a person's identity, and explicitly record their computational and editorial provenance. This distinction is particularly important in historical corpora, where available metadata is often incomplete, ambiguous, or shaped by historical biases.

Elements Even though the TEI standard offers both `<gender>` and `<sex>` elements, we deliberately use the `<gender>` element in our data model, since our classifications reflect socially and historically informed name-gender associations rather than biological characteristics.

Our pipeline internally distinguishes four output categories: `male`, `female`, `unknown`, and `institution`. The binary classification into `male` and `female` reflects the limited scope of what name-based methods can reliably infer: statistical associations between historical name forms and socially assigned gender categories. This enables an approximation of female representation in historically male-dominated corpora, but necessarily limits what can be captured beyond the binary (see Section 4). Entries the pipeline cannot resolve are assigned to `unknown` or `institution`, which is specific to the corpus of the use case. While TEI itself does not prescribe a fixed taxonomy for gender values, this deliberately small controlled vocabulary ensures interoperability with existing editorial workflows and authority data sources.

The `<gender>` element is encoded as a child of `<person>` rather than as the shorthand `@gender` attribute, because only an element allows the annotation itself to carry provenance, responsibility, and certainty metadata. The element-based approach furthermore permits future extensions such as temporal validity ranges, source references, alternative readings, or additional gender categories. The element is inserted after the last `<persName>` and before any `<birth>`, `<death>`, or `<note>` children.

```
<person xml:id="S0001126">
   <persName type="reg">
      <surname>Arnold</surname>
      <forename/>
   </persName>
   <gender value="female" evidence="assigned"
           resp="#gendertagger" cert="unknown"/>
   <note>Lehrerin in
        <placeName key="S0000357">Gnadenfrei</placeName>
   </note>
</person>
```

In this example, *gendertagger* adds only the `<gender>` element; all other markup, including the place reference, is taken unchanged from the *schleiermacher digital* register. The key `S0000357` identifies the edition's register entry for Gnadenfrei, which links the place to the GeoNames gazetteer.[5] The note reads *Lehrerin in Gnadenfrei* [female teacher in Gnadenfrei].

[5] https://schleiermacher-digital.de/S0000357 and https://www.geonames.org/3089025. Accessed 23 September 2026.

Note that the empty <forename/> combined with cert="unknown" illustrates the pipeline operating under uncertainty: here, the occupational context (*Lehrerin* [female teacher]) provides the basis for classification where the name record alone would not suffice; as only the LLM can use this context, no second method confirms the label.

Attributes The four attributes carry the editorial semantics of the annotation:

@value The attribute takes one of the four possible classification values: male or female, unknown or institution. The data model allows future additions.

@evidence Fixed to assigned, indicating that the annotation represents a computationally or historically assigned category rather than a self-reported or directly observed one. The attribute is encoded explicitly to distinguish the evidential basis from potential future evidence types.

@resp The entity responsible for the annotation: #gendertagger for entries produced by the automated pipeline, and a human-editor identifier (e. g. #abc) for entries that were verified or corrected through the review interface. The same TEI mechanism for responsibility attribution is thus applied to both human and machine-generated annotations.

@cert We restrict the TEI @cert attribute to three of the standard TEI values: high, low, and unknown. Human-reviewed entries are always marked as high. For automated entries the value is derived from the pipeline certainty tier produced in Section 2.5: unanimous or strong agreement across multiple methods, as well as cases decided by an authority signal, map to high; weak agreement maps to low; no consensus, a single voting method, or no usable signal maps to unknown. The value qualifies the resulting annotation, not the confidence of an individual method; unknown follows the TEI definition that the encoder does not wish to assert an opinion [TEI26, datatype teidata.certainty].

Certainty Scale TEI also allows medium and numerical probabilities between 0 and 1 for @cert [TEI26, datatype teidata.probCert]. We did not use numerical values because the pipeline has no calibrated probability to report: per-method confidences are not comparable, and the agreement score measures consensus, not correctness. We did not use medium because the pipeline distinguishes only two levels of agreement; like the ediarum framework, which offers only low and high for uncertain readings [TEL24], we keep the scale coarse. The finer tiers and agreement scores remain available in the pipeline output.

The element <gender> and the attributes @cert and @resp are standard TEI constructs [TEI26]. The TEI specification for gender is currently under active revision [BVBJ24], and our modeling choices are designed to remain compatible with its evolving recommendations. We apply these constructs to our use case in order to make the resulting data queryable and reviewable within existing editorial workflows, enabling downstream queries such as *all entries classified as female with* cert="high" or *all uncertain entries requiring human review*.

2.3 Workflow Description

The implemented pipeline arranges its workflow in three stages, each producing artifacts that the next stage consumes. In a first step the source TEI/XML register is flattened into a single table with one row per person and the relevant fields per entry: identifier, surname and forename(s), vital dates, free-text biographical note, alternative name forms, and, where present, a GND identifier. Entries that provide such an identifier are additionally resolved against the authority record, yielding a partial authority-based reference set used both as a classifier in its own right and as the evaluation reference for the other methods. Since authority data may itself contain errors or historically contingent assumptions, this reference set is used as a benchmark rather than as unquestioned ground truth.

Each gender-classification method is then applied independently to the prepared data and produces, per entry, a label drawn from a controlled vocabulary (`male`, `female`, `unknown`, with `institution` where the method can express it) together with a method-specific confidence score. The methods were kept independent instead of chaining them, so each can be re-run, replaced or removed without affecting the whole pipeline, and slow methods need only run once before being re-aggregated freely.

Per-method labels are joined into a single table from which a resulting label, the number of methods that returned a substantive answer, an inter-method agreement score, and a categorical certainty tier are derived. The resulting label is the automated output of aggregation, including any authority override, before human review. Agreement is quantified at two levels: pairwise between methods and per record. A lightweight review interface lets an editor verify or correct the entries flagged as low-certainty, and the verified annotations are folded back into the TEI source with provenance distinguishing human-reviewed from automated decisions.

2.4 Classifiers and Methods Overview

The enrichment pipeline combines five methodological families of automated gender classification, instantiated in the current configuration by ten components: the GND authority lookup, the parenthetical-qualifier rule, three dictionary-based classifiers (*gender-guesser*, *Wiki-Gendersort*, *HIVOTO*), two probabilistic classifiers (*nomquamgender* and the experimental *HIVOTO-XGBoost* model), two transformer-based classifiers, and a locally hosted large language model. Nine of these enter the reported aggregation; the experimental *HIVOTO-XGBoost* model is held out and reported only as a methodological extension. Each method has its own strengths and weaknesses, which motivates the use of a method ensemble and aggregation of the received results.

2.4.1 Authority-Data Lookup

GND The GND is widely used in German-speaking digital scholarship for person identification and can therefore be considered the classical approach. Linking entries to GND identifiers requires manual annotation and curation, which is a time-consuming process carried out by the editors of the edition. For entries that carry a GND identifier, we retrieve the corresponding gender information from the record. Because it retrieves a curated assignment, it is the most trustworthy single signal in the pipeline and used as a reference against which the other methods are evaluated. Unfortunately authority files can cover only parts of the historical data at hand, es-

pecially for regionally specific projects. In our case only 27.2% of entries in the *schleiermacher digital* register contain a GND-reference. Instead of querying the GND live during classification, we resolve identifiers against a local dump of the authority file.[6] This removes the pipeline's dependency on external service availability, and eliminates rate-limit and latency concerns. The trade-off is that the local snapshot must be refreshed to incorporate subsequent additions and corrections to the GND. The current backend retains source dumps and derived lookup snapshots as immutable artifacts, recording their checksums and the source data and compiler version used to produce each lookup. Each method execution references the exact lookup snapshot used, preserving traceability across updates. The choice of GND was driven by the predominantly German-language source material and the existing editorial authority links, which primarily use GND identifiers. The Virtual International Authority File (VIAF) was excluded because of ongoing discussion about reliability and service governance following its January 2025 redesign (see, for example, Bargioni [Bar25]). Future extensions could include usage of records in Wikidata and Getty's Union List of Artist Names (ULAN).

Parenthetical-qualifier rule This heuristic is implemented within *gendertagger* and scans the surname, forename, and biographical note for German kinship markers (e. g. *Frau des* … [wife of …], *Schwester von* … [sister of …], *Sohn der* … [son of …]) and structural disambiguators enclosed in parentheses and assigns `female` or `male` accordingly; if no marker is found, the rule abstains. Unlike the name-based classifiers, it does not rely on a name dictionary but on textual context, making it complementary to methods that require a usable forename. The rule is edition-specific: it was designed for naming conventions found in the register data, where women are frequently identified through their relationship to a male principal rather than by name. Since the rule targets information added by the human editors, together with the GND-Lookup, it acts as an authority override in the aggregation step.

2.4.2 Dictionary-Based Name Lookups

Three methods relying on fixed dictionaries of classified names were used in this study. The methods provide a fast and deterministic classification, relying only on the forename of a person, but carry the biases and limitations of the datasets used for their creation.

gender-guesser An open-source Python package built around a large multilingual first-name dictionary (`nam_dict.txt` by Jörg Michael), returning one of six labels (`male`, `mostly_male`, `andy`, `mostly_female`, `female`, `unknown`). Multi-token forenames are resolved to a single label by token consensus. The underlying dictionary seems to be no longer actively maintained. [Pé16]

Wiki-Gendersort A name-to-gender mapping derived from English Wikipedia biographies by counting gendered pronouns in article introductions, distributed as a flat-file lookup of 694,376 entries with five labels (`M`, `F`, `UNI`, `INI`, `UNK`). [Bé20, BGSL20]

[6] The dump was created in March 2026.

HIVOTO The *Historisches Vornamentool*, is a German-language historical first-name database covering roughly 202,000 spelling variants consolidated into approximately 17,800 normalized forms, each with a sex-category assignment and an indicator for ambiguity. The variants are drawn from handwritten and printed sources of the 17th to mid-20th century. Of the dictionary-based methods in our roster, HIVOTO is the only one specifically targeted at historical German name variation, and is therefore the most directly applicable name-based reference for the *schleiermacher digital* corpus. [Lie26]

2.4.3 Probabilistic Classifiers

Contrary to the dictionary-based methods, these methods return a probability that can be thresholded while preserving uncertainty information. In their application they also are limited by the coverage of their underlying datasets.

nomquamgender (NQG) An open-source probabilistic classifier built on a reference corpus aggregating 36 name–gender data sources across 150 countries and more than a century. Per name, it returns the probability $p(\mathrm{gf})$ that the name has been used for a person "gendered female"; we threshold the probability at $p = 0.5$ into a categorical label and retain p as the confidence score. [BCL23]

HIVOTO-XGBoost As an experimental extension, we trained a character-n-gram XGBoost model on the HIVOTO database, using 118,286 fully written-out name forms with binary labels (93,686 male / 24,600 female). On a stratified 15% held-out set the model reaches 97.3% accuracy with per-class F1 of 0.98 (male) and 0.94 (female). The model is excluded from the reported aggregation and reserved for future evaluation; full training parameters are documented in the source repository.

2.4.4 Transformer-Based Name Classifiers

We also used two pretrained transformer-based classification models found on the model hub HuggingFace.[7] Both models illustrate a recurring transparency problem in community-published machine-learning models. Central information essential for assessing their suitability in scholarly contexts, including training data provenance, language and demographic coverage or meaningful evaluation is often omitted from the information published alongside the model.

DistilBERT (`padmajabfrl/Gender-Classification`) A DistilBERT model trained for binary gender classification. The author provides no further information than:

> "This model is a fine-tuned version of `distilbert-base-uncased` on an unknown dataset." [pad25]

[7] https://huggingface.co. Accessed 23 September 2026.

BERT-genderize (`imranali291/genderize`) A BERT model fine-tuned for binary M/F classification on the `Dannel/gender-name` corpus[8], a list of ∼178,000 first names. The training data is predominantly Spanish-language with a strong Hispanic skew. The model was trained with a learning rate of 2×10^{-5} over 20,005 steps and a maximum input length of 32 tokens. [Ali25]

2.4.5 Locally Hosted Large Language Model

gpt-oss:20b We used an open-weight 20-billion-parameter LLM run locally via Ollama under the model tag `gpt-oss:20b`. Unlike all other methods, it is given the full register entry: surname, forename(s), vital dates, biographical note, alternative name forms. It is then used to return a structured JSON object with a label drawn from {`male`, `female`, `organisation`, `unknown`}, a confidence value, and a free-text reason. The biographical context lets it resolve cases that defeat the name-only methods[9] (initials with disambiguating dates, institutional names, non-German forms identified through their note text) [Ope25]. However, these contextual descriptions are themselves the result of prior editorial research and interpretation, meaning that existing editorial biases may already be embedded in the source material and subsequently reproduced through the model's classifications. The LLM result should therefore not be read as evidence that the model has superior name-based gender knowledge.

2.5 Aggregation Strategy

The methods are run in parallel and their per-method labels are combined in three explicit steps: normalization, voting (with two authority overrides), and certainty assignment. Of the ten implemented components, nine enter the reported aggregation; the experimental *HIVOTO-XGBoost* model is held out. Of those nine, two (the parenthetical-qualifier rule and the GND authority lookup) function as authority overrides layered on top of the vote, while the remaining seven methods enter the unweighted majority vote. The per-method confidence values are recorded but do not influence the vote.

2.5.1 Normalization

Unfortunately the methods do not share a common label vocabulary. *gender-guesser* distinguishes `male`/`mostly_male`/`andy`, *HIVOTO* returns a German-language certainty indicator alongside the label, the LLM emits `organisation`, and the statistical models return probabilities. Before voting, every per-method label is mapped onto a shared four-value alphabet {`male`, `female`, `institution`, `unknown`}: leaning labels (`mostly_male`, `mostly_female`) collapse to their definite counterpart, all genuinely ambiguous outputs (`andy`, `conflict`, `uncertain`) collapse to `unknown`, and absent or unparseable answers are recorded as abstentions rather than as `unknown` votes. The distinction is critical because an abstention means the method had no input to work with (e. g. an entry without a forename), while `unknown` means ambiguities in the input, because of which the method could not decide on a label.

[8] https://huggingface.co/datasets/Dannel/gender-name. Accessed 23 September 2026.
[9] See the example person entry in Section 2.2.

2.5.2 Voting

For each entry the aggregator applies a fixed priority chain: parenthetical-qualifier rule, then GND lookup, then the majority vote across the seven pooled methods. The parenthetical-qualifier rule scans the surname, forename, and biographical note for German kinship markers and structural disambiguators in parentheses; when it fires, its label is taken as the resulting label and the entry is promoted to the certainty tier `high`. Otherwise, a non-`unknown` GND lookup result is taken as the resulting label and the entry is likewise promoted to `high`. When neither authority signal fires, the resulting label is the plain (unweighted) majority of the substantive votes across the seven pooled methods that did not abstain, with one tie-breaking rule: when `unknown` ties with `male`, `female`, or `institution`, the substantive label wins, since `unknown` encodes "no information" rather than positive evidence against the substantive option. The agreement score recorded for the entry is the share of votes carried by the winning label among the methods that did vote; for entries decided by an authority override it is recorded as the share of the remaining methods that agree with the override label.

This priority ordering reflects a pragmatic ranking of evidence types: the parenthetical-qualifier rule encodes information explicitly inserted by the human editors (e. g. *Tochter des* . . . [daughter of . . .]), and the GND lookup retrieves a curated authority assignment. Both are therefore treated as higher-priority signals than purely name-based statistical classifiers. The trade-off is that errors in high-priority signals may propagate into the resulting label unless they are caught by a higher-priority rule, strong disagreement among the remaining methods, or subsequent editorial review.

2.5.3 Certainty assignment

From the agreement score and the number of voting methods, each entry is assigned a categorical pipeline certainty tier: `high` when at least two methods voted and at least three quarters of them agreed (or whenever an authority signal fired); `low` when at least two voted and a majority of half or more agreed; `contested` when at least two voted but agreement fell below half or a tie between substantive labels remained unresolved; `single` when only one method had useful output and that lone method was the LLM with self-reported `high` certainty; `uncertain` when the lone method was the LLM with `low` or `uncertain` self-reported certainty; and `no_data` when no method could vote. These pipeline tiers are recorded internally; how they are projected onto the TEI `@cert` attribute is described in Section 2.2.

2.5.4 Inter-method agreement

Beyond the resulting label and certainty tier, the pipeline records inter-method agreement at two levels. Pairwise agreement between methods is measured using Cohen's κ, restricted to entries on which both methods returned a substantive (non-abstention) answer. This identifies which methods behave similarly and could be dropped in subsequent configurations. Per-record agreement is quantified as the share of answering methods that voted for the winning label, and each entry is sorted into one of six tiers: `unanimous`, `strong_majority` ($\geq 75\%$), `weak_majority` ($\geq 50\%$), `no_consensus` ($< 50\%$), `single_rater` (only one method committed), and `no_answer`.

The underlying assumption is that if multiple methods using different technical approaches and different training data converge on the same label, the classification is more likely correct; conversely, disagreement becomes the primary triage signal that directs human attention to entries where automated annotation is least reliable.

3 Case Study: schleiermacher digital

The development of *gendertagger* was conducted in collaboration with the *schleiermacher digital* edition,[10] which provides digital access to the correspondence, lectures, and diaries of the theologian Friedrich Schleiermacher (1768–1834), as well as additional contextual material. As a long-term digital scholarly editing project, the edition brings together heterogeneous textual sources that contain a wide range of person references and contextual information. This makes it particularly suitable for investigating the gender data gap in historical person registers, as it enables systematic analysis of how individuals are represented, recorded, and potentially underrepresented in structured scholarly datasets.

3.1 The Person Register

The edition's person register contains 8,277 entries at the time of writing. Only 2,253 of those entries (27.2%) carry a GND identifier resolvable to a gendered authority record; the remaining 6,024 entries (72.8%) require alternative classification approaches. This distribution is representative of the practical limits of authority-file-based enrichment for a register of this size and historical scope: even a relatively well-curated GND attachment leaves nearly three quarters of the entries without an authoritative gender signal, motivating the multi-method approach used in this study.

3.2 Coverage per Method

Coverage on the *schleiermacher digital* register varies systematically with each method's input requirements (Table 1). The local LLM, which receives the full register entry including biographical note, is the only method that produces a label for every entry. The five name-based classifiers (*gender-guesser*, the two transformers, *HIVOTO*, and *nomquamgender*) each cover the same 6,134 entries (74.1%): the gap to full coverage is exactly the set of entries with no usable forename. Wiki-Gendersort covers a slightly smaller subset (73.4%). The parenthetical-qualifier rule reaches 33.5% of entries. In these limited cases, the register provides additional biographical information, such as e. g. *Frau des* … [wife of …] or *Schwester von* … [sister of …].

3.3 Per-method label distributions

While coverage tells us how often each method commits to *some* label; the per-method label distribution (Table 2) shows *which* labels each method tends to assign and can expose systematic

[10] https://schleiermacher-digital.de

Method	Coverage	Share
Local LLM (gpt-oss:20b)	8,277	100.0%
gender-guesser	6,134	74.1%
HIVOTO	6,134	74.1%
HF DistilBERT (padmajabfrl)	6,134	74.1%
HF BERT-genderize (imranali291)	6,134	74.1%
nomquamgender	6,134	74.1%
Wiki-Gendersort	6,075	73.4%
Parenthetical-qualifier rule	2,771	33.5%
GND authority lookup	2,253	27.2%

Table 1: Per-method coverage on the *schleiermacher digital* person register ($N = 8{,}277$). Coverage is the number of entries on which the method returned a substantive label (not an abstention).

Method	male	female	inst.	unk.	abst.
GND	2,048	205	0	0	6,024
Parenth. qualifier	1,133	1,159	7	472	5,506
gender-guesser	4,332	1,278	0	524	2,143
Wiki-Gendersort	4,457	1,292	0	326	2,202
HIVOTO	4,413	1,457	0	264	2,143
nomquamgender	4,379	1,380	0	375	2,143
HF BERT-gz	4,653	926	0	555	2,143
HF DistilBERT	4,569	1,117	0	448	2,143
LLM	5,426	1,897	119	835	0
Resulting label	5,452	1,867	83	875	0

Table 2: Per-method label distributions on the *schleiermacher digital* register ($N = 8{,}277$). The resulting label row shows the automated output of the aggregation pipeline before manual review.

bias profiles. Three patterns are visible in the analysis. First, the GND-matched subset is itself strongly skewed: of the 2,253 entries with a GND reference, 90.9% resolve to `male` and only 9.1% to `female`. Second, the two transformer-based models diverge markedly on the same 6,134-entry coverage set: HF BERT-genderize predicts $\sim$5$\times$ as many `male` as `female` entries (56.2% vs. 11.2%), while HIVOTO on the identical input predicts only $\sim$3$\times$ as many (53.3% vs. 17.6%). The bias of the Genderize-model toward `male` is the dominant source of its low κ against the LLM (Figure 1). Third, the parenthetical-qualifier rule is the only method whose label distribution is approximately gender-balanced (13.7% `male` vs. 14.0% `female`).

3.4 Per-method Accuracy against GND

Using the 2,253 entries with a gendered GND authority record as a benchmark set allows a direct comparison of per-method classification quality (Table 3). Effective accuracy, correct labels divided by the full benchmark size (abstentions count against the method), ranges from 86.5% (the two transformer classifiers, which abstain on $\sim$12% of the benchmark) to 99.7% (the local LLM, which abstains on only 3 entries). An imbalance shows between male and female recall. The two transformer classifiers achieve 88–89% on male names but only 60–

71% on female names, the largest per-class gaps in the cohort; gender-guesser, Wiki-Gendersort, HIVOTO, and nomquamgender close the gap considerably (84–91% female recall against 88–90% male); only the LLM is balanced across classes (99.7% male, 100% female on the GND benchmark). The benchmark itself is small and male-dominated (2,048 male vs. 205 female), so the results should be interpreted with appropriate caution. A further asymmetry concerns the LLM specifically: unlike the name-only classifiers, it receives the full `note` field, which frequently contains biographical prose. A keyword probe finds overt gender signals in 29% of benchmark entries, but with a strong gendered bias. 94% of female entries carry a class-matching signal (*Tochter* [daughter], *Frau des …* [wife of …], occupational feminines), against only 21% of male entries. So the LLM's perfect female recall could have its foundation largely in context reading, while its male recall rests predominantly on the name string.

	Answered		Correct		Eff. acc.	Recall (%) with Wilson 95% CI			
Method	*n*	%	*n*	%	%	male	95% CI	female	95% CI
gender-guesser	1,987	88.2	1,978	99.5	87.8	88.1	[86.6, 89.4]	84.9	[79.3, 89.1]
Wiki-Gendersort	2,005	89.0	1,994	99.5	88.5	88.9	[87.5, 90.2]	84.9	[79.3, 89.1]
HIVOTO	2,000	88.8	1,999	100.0	88.7	88.5	[87.0, 89.8]	91.2	[86.5, 94.4]
nomquamgender	2,028	90.0	2,027	100.0	89.9	90.1	[88.8, 91.4]	88.3	[83.2, 92.0]
HF DistilBERT (padmajabfrl)	1,976	87.7	1,948	98.6	86.5	88.0	[86.6, 89.4]	70.7	[64.2, 76.5]
HF BERT-genderize (imranali291)	1,982	88.0	1,948	98.3	86.5	89.1	[87.7, 90.4]	60.0	[53.2, 66.5]
Local LLM (gpt-oss:20b)	2,250	99.9	2,247	99.9	99.7	99.7	[99.4, 99.9]	100.0	[98.2, 100.0]

Table 3: Per-method performance on the GND benchmark subset ($N = 2{,}253$; 2,048 male, 205 female). *Answered* counts entries on which the method committed to `male` or `female` (the complement are abstentions or ambiguous outputs); *Correct* % is the share of correct labels among answered entries; *Effective accuracy* divides correct labels by the full benchmark size, so abstentions count against the method. Per-class recall is correct predictions divided by the GND class size and is reported with Wilson 95% confidence intervals; the female-class intervals are wide ($n = 205$) and should temper any per-method ranking.

3.5 Where methods agree and disagree

3.5.1 Between Methods

Pairwise agreement (Cohen's κ) across the eight classifiers for which pairwise coverage overlap is sufficient reveals three clusters (Figure 1): the HIVOTO dictionary with nomquamgender, the dictionary methods gender-guesser and Wiki-Gendersort, and the two transformer classifiers, while the LLM sits apart from all three. Because the LLM is the only classifier reasoning from biographical context rather than from name strings alone, it resolves entries (initials, institutional names, non-German forms) that the name-only methods cannot process. The bias of the Genderize-model toward `male` is the dominant source of its low κ against the LLM.

3.5.2 Per Record

Per-record agreement is quantified as the share of answering methods that voted for the winning label. The distribution is heavily right-skewed among entries for which more than one method

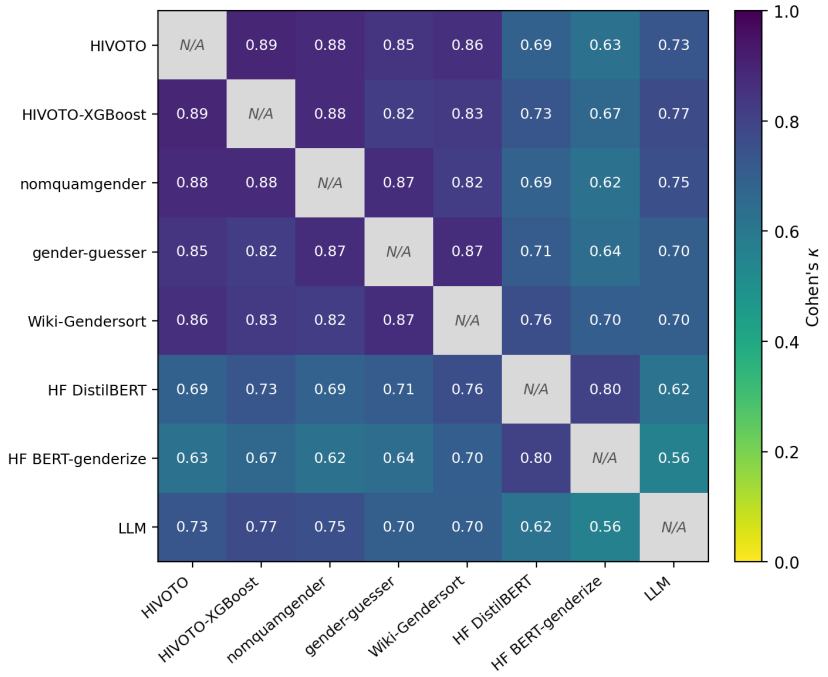


Figure 1: Pairwise inter-method agreement (Cohen's κ) on the *schleiermacher digital* register. GND and the parenthetical-qualifier rule are excluded because their coverage overlap with the name-based methods is too small for a stable pairwise estimate.

produced a usable answer. Of these 6,134 entries, 4,441 (72.4%) are unanimous, 843 (13.7%) reach a strong majority, 488 (8.0%) reach only a weak majority, and 362 (5.9%) show no consensus. The remaining 2,143 entries fall into the `single_rater` or `no_answer` categories and are therefore excluded from Table 4. The breakdown by majority label reveals a systematic asymmetry: the methods agree unanimously on 80.8% of the `male` entries but only on 56.8% of the `female` entries, suggesting that `female` is both smaller in absolute terms and harder to assign reliably. This is in part because women in this corpus are disproportionately referenced through relational descriptors that name-based methods cannot resolve.

Majority label	N	unanimous	strong_maj.	weak_maj.	no_consensus
`male`	4,548	3,677 (80.8%)	524 (11.5%)	216 (4.7%)	131 (2.9%)
`female`	1,346	764 (56.8%)	314 (23.3%)	213 (15.8%)	55 (4.1%)
`ambiguous`	240	0 (0.0%)	5 (2.1%)	59 (24.6%)	176 (73.3%)

Table 4: Per-record agreement tier for entries with more than one usable method response.

These agreement tiers map onto the pipeline certainty tiers, but not one-to-one. `Unanimous` and `strong_majority` entries, as well as any entry where an authority override fired, are promoted to `high`. `Weak_majority` maps to `low`, `no_consensus` to `contested`. The

Pipeline certainty tier	TEI `@cert`	Entries	Share
`high`	`high`	6,934	83.8%
`single`	`unknown`	710	8.6%
`uncertain`	`unknown`	338	4.1%
`low`	`low`	257	3.1%
`contested`	`unknown`	38	0.5%

Table 5: Pipeline certainty tiers before review and their TEI `@cert` value ($N = 8{,}277$). Human-reviewed annotations are exported as `high`.

`single_rater` tier splits depending on the LLM's self-reported certainty: entries where the LLM committed with high certainty become `single`, those with low or uncertain certainty become `uncertain`. On the *schleiermacher digital* register, the resulting distribution is heavily concentrated at `high` (Table 5): 6,934 entries (83.8%) reach the tier `high`, while only 295 (3.6%) fall into the genuinely problematic `low` or `contested` tiers. The remaining 1,048 entries (12.7%) are LLM-only cases (`single` or `uncertain`), entries on which only the LLM was able to commit to a label.[11]

Manual inspection of the most-contested records shows that disagreement clusters around four recurring profiles: relational naming conventions where the entry's forename field contains the name of a related person rather than the individual themselves; title-only entries without a usable forename; initials-only entries where dictionary methods have nothing to match against; and institutional names mis-parsed as personal names (Table 6).

Many of these disagreements stem from the way information is recorded in the register itself. Some entities are only listed by their relation to another entity, with the relation encoded in parentheses, e. g. *(Frau des **Heinrich**)* [wife of Heinrich], *(Mutter der **Antonie**)* [mother of Antonie], *(Mann der **Marianne**)* [husband of Marianne], as the entry's own forename. This caused the strictly forename-based methods to mislabel the entries and was addressed using the parenthetical-qualifier rule during aggregation.

3.5.3 Errors in Authority Data

In a very small number of cases the input data seemingly contained errors (Table 7). These issues surfaced only at the aggregation step and helped validate the multi-method approach. In the first case, `S9145415`, the register entry describes Auguste Ernestine Aemilie Klein, daughter of Ernst Ferdinand Klein, but its GND ID points to an unrelated record (GND 1165816261, "Klein, Auguste", variant "Klein, August", a Berlin painter born 1777). The dictionary methods, the parenthetical-qualifier rule (firing on *Tochter des Ernst Ferdinand Klein* [daughter of Ernst Ferdinand Klein]), and the LLM all return `female`; because the parenthetical-qualifier rule sits above GND in the aggregation chain, the resulting label becomes `female` with the certainty tier `high` and the mis-pointing GND label is silently corrected. The other two cases are errors in the editorial biographical notes that mislead the LLM. In `S3729958`, the note "Portrait- und Genremalerin in Berlin" [portrait and genre painter in Berlin (feminine form)] should read *Genremaler* [genre painter (masculine form)]; the feminine suffix *-in* cues the LLM to `female`.

[11] The five tiers are mutually exclusive and exhaustive: $6{,}934 + 710 + 338 + 257 + 38 = 8{,}277$.

xml:id	Entry / note	Vote pattern	Resulting label
S0001932	*Frölich, (Frau des Heinrich)* [wife of Heinrich] "Frau des Heinrich Frölich" [wife of Heinrich Frölich]	Four of six name-based methods read "Heinrich" as a forename and return male (HFGC, HFG, HV, WGS); the parenthetical-qualifier rule fires on "Frau des" [wife of] → female; LLM → female from the note.	female (tier high; rule overrides the name-based cluster)
S0001502	*Canitz, Baronin von* [Baroness von Canitz] (no biographical note)	Title-only entry without forename; the name-based methods scatter (HFGC, HFG → male; NQG → female; GG, HV, WGS ambiguous); LLM → female via German title morphology (*Baronin* [baroness]).	unknown (tier contested)
S0001317	*Bing, A. H.* "Arzt in Berlin" [physician in Berlin]	Six of nine methods abstain or return ambiguous because "A. H." is unparseable as a forename; only HFGC, HFG, and the LLM commit, all to male, the LLM via the masculine "Arzt" [physician].	male (tier low; three male, three unknown votes)
S9437075	*Fournier, L. und Compagnie* [Fournier, L. and Company] "wohnhaft in der Friedrichsgracht 50" [residing at Friedrichsgracht 50]	Trading firm; name-based methods read "L." as a forename and split (HFGC → female; HFG → male; rest ambiguous); only the LLM flags the entry as an institution.	unknown (tier low)

Table 6: Examples for the four dominant disagreement types ($N = 850$ entries with a weak majority or no consensus). Vote patterns are taken from the per-method results; resulting labels are shown before review.

In S0003113, a missing space concatenates *Prediger in Halle* [preacher in Halle] into *PredigerinHalle*, creating the apparent word *Predigerin* [female preacher] and producing the same misreading. In both cases the name-based methods resist, the majority outvotes the LLM, and the resulting label remains male.

3.6 Resulting Gender Distribution

After manual review of low-agreement cases, the final reviewed classification yields a strongly imbalanced distribution (Figure 2): of the 8,277 entries, 5,562 (67.2%) are labeled male, 1,932 (23.3%) female, 120 (1.4%) institution, and 663 (8.0%) remain unknown. The male-to-female ratio of roughly 2.9:1 is itself a finding about the source rather than about the method: the private and academic correspondence of an early-nineteenth-century Berlin theologian skews structurally toward male correspondents (university colleagues, ecclesiastical and political officeholders, publishers). The agreement evidence corroborates this: the methods agree unanimously on 91.5% of the male entries in the final reviewed output but only on 58.5% of the female entries (in Table 4 we see the same asymmetry at 80.8% and 56.8% over the voting-eligible subset).

xml:id	Name	GND	LLM	Resulting label	Verdict
S9145415	Klein, Auguste Ernestine Aemilie (“Tochter des Ernst Ferdinand Klein”) [daughter of Ernst Ferdinand Klein]	male	female	female	Register ref links to an unrelated GND record (a male painter named Klein, Auguste/August)
S3729958	Magnus, Leopold (“Portrait- und Genremalerin in Berlin” [portrait and genre painter in Berlin (feminine form)])	male	female	male	Note erroneously feminizes *Maler* [painter (masculine form)] as *Malerin* [painter (feminine form)]
S0003113	O’Bern, Marcus Phillip Ludwig de (“französisch-reformierterPredigerinHalle” [intended reading: French Reformed preacher in Halle])	male	female	male	LLM was misled by an error from concatenation: *Prediger in* [preacher in] as *Predigerin* [female preacher]

Table 7: Three cases of errors in data sources deemed authoritative.

4 Limitations and Reproducibility

4.1 Limitations

Normalization and Binary Scope In the final output, we collapse all classifier results into a binary scheme plus institution and unknown. This design choice reflects the constraints of name-based inference on historical datasets and the structure of the available source material. While this enables the large-scale measurement of female representation in historically male-dominated corpora, it reproduces the categorical limitations embedded in the data. As a consequence, forms of gendered difference that are not encoded in historical naming conventions remain outside the scope of the classifier and are either mapped to unknown or absorbed into the binary categories. Since we record all intermediate classifier outputs, further analysis of cases classified as conflicting or ambiguous is still possible and could provide additional insights and open up new research questions.

Single-corpus Evaluation All numbers reported here are for one nineteenth-century German register; transferability to corpora with different languages or temporal and cultural backgrounds was not evaluated.

Authority Bias *GND*, *HIVOTO*, and the editorial notes carry historical and curation biases that the pipeline inherits and partially propagates (see Errors in Authority Data).

Model Bias The two transformer classifiers were trained on undocumented or non-German data and exhibit strong male bias; we keep them in the ensemble as a transparency case rather than because they improve aggregate quality. Besides *HIVOTO*, no method explicitly targets historical data.

Unweighted aggregation Per-method confidences are recorded but do not influence the vote; ablations against confidence-weighted voting are not reported.

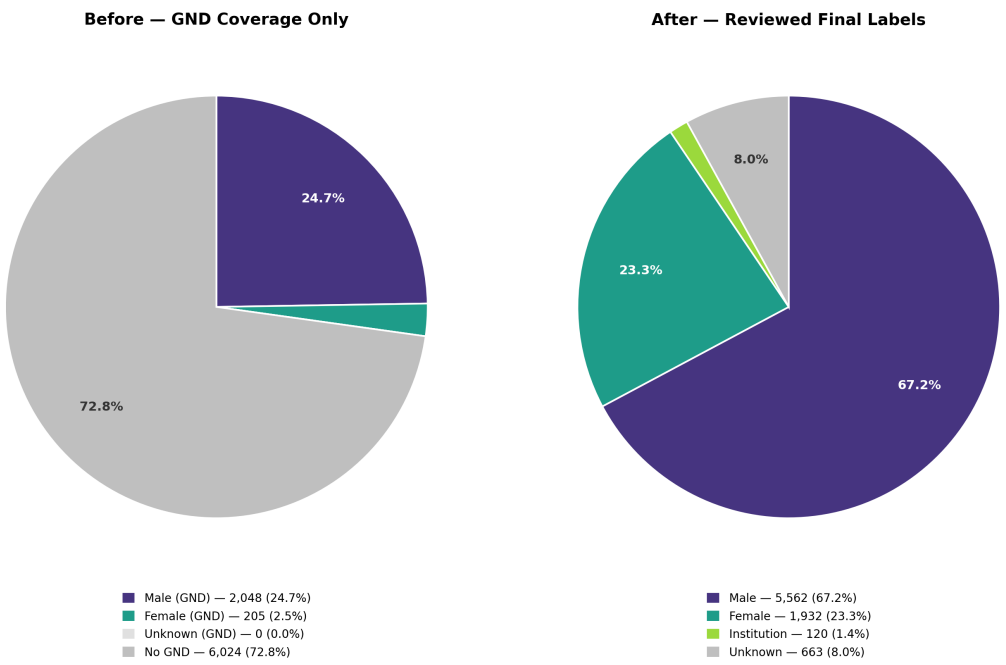


Figure 2: Gender coverage of the *Schleiermacher* person register ($N = 8{,}277$), before (left) and after (right) the aggregation-and-review pipeline.

4.2 Reproducibility

To support reproduction we record the operational details that most directly affect the published numbers. The pipeline is implemented in Python and run locally end-to-end. The LLM is `gpt-oss:20b` [Ope25] served by Ollama; the prompt is a fixed system message instructing structured-JSON output conforming to the following schema: {`gender`∈{`male`, `female`, `organisation`, `unknown`}, `cert`∈{`high`, `low`, `uncertain`}, `confidence`∈[0,1], `reasoning`}, with up to three validation-failure retries. The `cert` field holds the LLM's self-reported certainty (Section 2.5), not a TEI `@cert` value. The user message contains the available register fields (`surname`, `forename`, `birth`, `death`, `note`, `alt_names`). The transformer classifiers are `padmajabfrl/Gender-Classification` and `imranali291/genderize` (HuggingFace, default revisions at run time); the *NQG* threshold is $p = 0.5$; *HIVOTO-XGBoost* uses a fixed RNG seed and a stratified 15% held-out set used only for early stopping. The *GND*, *HIVOTO*, and *schleiermacher digital* snapshots are dated to the case study run (March 2026). The full prompt text, schema, model identifiers, package versions, are distributed with the source repository [KNW26].

5 Discussion

The editorial review of the case study provides a first check of the certainty tiers. Editors reviewed all 633 entries with the tiers `low`, `contested`, and `uncertain`. They changed an existing label in 5.8% of the `low` entries but in 26.3% of the `contested` entries; most other corrections supplied a label where the pipeline had returned `unknown`. In the `low` tier, 14 of the 15 changed labels were corrected to `female`. Two groups remained unreviewed: all `high` entries and all but one of the 710 `single` entries. For `high`, the GND benchmark provides only an indirect estimate; the `single` entries rest on an LLM whose accuracy on the GND benchmark partly derives from the biographical notes.

From a research software engineering perspective, the individual classifiers are the most replaceable part of the workflow: they can be added, replaced, or held out, as with *HIVOTO-XGBoost*. What has to remain stable is the layer around them: normalization onto a shared label set, the distinction between abstention and `unknown`, and per-method provenance. External resources need the same care: the GND snapshot is versioned, whereas the transformer models were used at their default revisions.

The NFDI basic service TS4NFDI provides the provision, harmonization, and mapping of terminologies [BBF+25]. Through such a service, the mappings from each method's raw labels onto our label set could be published as reusable terminology mappings instead of remaining in the source code.

6 Conclusion

The case study suggests three transferable observations for editorial projects in a similar position. First, GND coverage alone is unlikely to suffice for any nineteenth-century or earlier register of comparable size; the long tail of non-authority entries demands a fallback strategy. Second, multi-method agreement provides not only a label but a meaningful triage signal: the $\sim$10% of entries on which the methods cannot agree concentrates exactly the cases where editorial attention is most warranted. Third, source-specific structural patterns — in this register, the relational naming convention for women — can be addressed by lightweight, edition-specific rules (the parenthetical-qualifier method here), provided the combination strategy is open enough to admit them as full peers of the off-the-shelf classifiers.

Our study demonstrates how semi-automated gender annotation can support metadata enrichment workflows in TEI/XML-based digital editions while preserving editorial oversight and interpretative responsibility. By combining multiple classification methods, integrating uncertainty-aware review mechanisms, and embedding annotation into existing editorial infrastructures, the system contributes to ongoing discussions on bias, metadata enrichment, and human-centered AI in the Digital Humanities.

The results indicate that no name-only classification method proved sufficiently reliable across the heterogeneous and historically ambiguous data of the register; the LLM's comparatively high accuracy is partly attributable to the editorial biographical notes it consumes (see Section 4). The combination of authority data, heuristic approaches, statistical models, transformer-based classifiers, and local language models produced more robust results when embedded in an uncertainty-

aware editorial workflow. Particularly valuable was not only the generation of gender annotations itself, but the ability to identify low-confidence and conflicting cases that required closer editorial attention. In this sense, the project demonstrates that semi-automated gender annotation is most effective not as a replacement for scholarly interpretation, but as a means of structuring and prioritizing editorial decision-making under conditions of uncertainty.

At the same time, the transferability of the approach depends on the linguistic, historical, and cultural characteristics of the respective corpus. Naming conventions, authority coverage, and gender associations vary substantially across periods and regions, meaning that edition-specific adaptation remains necessary. The modular and uncertainty-aware design of *gendertagger* is intended to support precisely such contextual adjustment rather than assuming universally stable gender classifications. We provide the work-in-progress code for reuse and adaptation [KNW26].

As geographer Joni Seager observed: "what gets counted counts" (cited in [DK20]). The same holds for digital editions: if gender is not represented as an analyzable category, structural imbalances remain difficult to detect and assess. By introducing *gendertagger* and integrating gender as a queryable attribute in editorial registers, we take a concrete step toward improving the analyzability of gendered patterns in digital scholarly data.

If gender is a documented category, researchers can, for instance, systematically ask which women were part of a literary network, how often they were mentioned compared to their male contemporaries, and in what roles they appeared. They can also trace whether women and other marginalized groups wrote about different topics or in different styles, and whether those differences reflect genuine creative choices or the constraints imposed on them by society. These are possible follow-up research questions that can only be pursued if gender is modeled in the first place.

But the implications reach beyond academic research. When gender becomes a legible category in our digital cultural heritage, it sends a signal about what we consider worth knowing. The case study demonstrates that even imperfect and uncertainty-aware gender annotations substantially expand the analytical possibilities of digital scholarly editions. It insists that the stories of women and other marginalized groups are not footnotes to history, but part of its fabric. And it gives future readers the tools to ask questions that previous generations simply could not.

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Data availability: The code for the project is under active development and publicly available [KNW26].

Declaration on the use of AI: Anthropic's Claude Opus models (versions 4.5–5) and Sonnet variants, OpenAI's GPT models (versions 5.1–5.5), and Google's Gemini Pro and Flash models,

[12] https://gender-data.bbaw.de/. Accessed 23 September 2026.
[13] https://www.bbaw.de/bbaw-digital/ki. Accessed 23 September 2026.
[14] https://schleiermacher-digital.de/S0010277. Accessed 23 September 2026.

including Gemini 3.5 Flash, were used for spelling, grammar, and phrasing suggestions during manuscript preparation and for code generation within agentic coding workflows during software development.

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